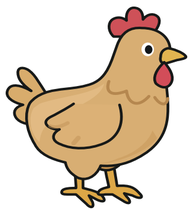
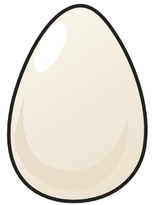
1 Defect understanding and report grouping

Defect descriptions (what each defect is)

Clustered reports (by underlying defect)



A chicken-and-egg problem



To group reports, we need to understand which defects they describe.

To understand defects, we compare reports within candidate groups.

Defect understanding guides grouping; grouping refines defect understanding.

2 Initial clustering is a hypothesis, not the final answer

Crowdsourced reports

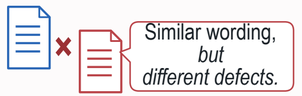
"payment failed ..."
"double charged ..."
"order cancel ..."
"refund not received ..."
...

Initial clustering (e.g., LLM)

But the grouping may still contain errors

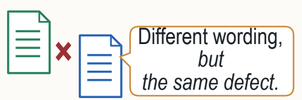
False merge

Reports from different defects are placed in the same group.



False split

Reports from the same defect are placed in different groups.



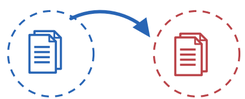
3 Iterative verification and refinement (FAVR)

1. Select pairs to verify
Find pairs whose estimated relation conflicts with the current grouping.

2. Verify with LLM
Obtain direct judgment SAME or DIFFERENT (with possible noise).

3. Refine the partition
Refine using accumulated direct judgments (correlation clustering).

4. Updated grouping changes the query context
The revised partition reveals new potential pairs to verify.



Verification guides refinement, refinement guides next verification.